

Introduction to Data: Summary and Visualization

STAT 223 – Applied Analytics

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Measures of Center

Measure of Center

A **measure of center** of data is a number that represents the “typical” value in the data.

Examples:

- mean
- median
- mode
- weighted mean
- mid-range

Sample Mean

Sample Mean

The **sample mean** is the center of gravity of the data.

It is the most commonly used measure of center, whether it should be used or not.

This is the value you think of when you think of the word “average”.

Sample Mean

We denote the sample mean as $\bar{x}$ (we say “x-bar”). We denote the true population mean μ (called “mu”, it’s a Greek lowercase m)

We tend to use Latin letters (Aa , Bb , Cc , Dd , ...) for *sample statistics*, and Greek letters ($A\alpha$, $B\beta$, $\Gamma\gamma$, $\Delta\delta$, ...) for *population parameters*.

Sample Mean

The formula is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Written out, this formula is:

$$\bar{x} = \frac{1}{n} (x_1 + x_2 + \cdots + x_{n-1} + x_n).$$

In English, this just means: *Add all your data values together and divide by how many there are.*

Sample Mean

Some notes on the sample mean:

- Every data value is included in the calculation.
- Every data value counts the same in the calculation.
- The order does not matter in the calculation.

One thing to think about is how each data value affects the measure of center. If a single data value can change the measure to an infinitely large value, then the measure is not **robust** to outliers.

The mean is *not* robust. A single outlier can make the mean incredibly large (or small). The next example illustrates this.

Sample Mean Example

Suppose I went to Innkeeper's Coffee and asked the customers there what they make in a year, to see what the mean salary of Innkeeper's Coffee customers is. Here are the salaries (in thousands of dollars):

32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1

By hand:

$$\begin{aligned}\bar{x} &= \frac{32.3 + 56.1 + 29.6 + 78.9 + 64.3 + 57.0 + 29.1}{7} \\ &= \frac{347.3}{7} \\ &= 49.614\end{aligned}$$

In R:

```
salaries <- c(32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1)
mean(salaries)
```

Sample Mean Example

Now suppose Elon Musk walks into the coffeeshop. We add his \$20,690,240,000 yearly income to the sample:

32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1, 20690240.0

By hand:

$$\begin{aligned}\bar{x} &= \frac{32.3 + 56.1 + 29.6 + 78.9 + 64.3 + 57.0 + 29.1 + 20690240.0}{8} \\ &= \frac{20690587}{8} \\ &= 2586323\end{aligned}$$

In R:

```
salaries2 <- c(32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1, 20690240.0)
mean(salaries2)
```

Sample Mean Example

Thus, with the inclusion of that single outlier, the mean increased by a factor of 10^5 .

Because of this, we can see that the mean is definitely *not* robust to outliers.

Weighted Mean

The **weighted mean** of the sample is similar to the regular mean, but each value in the sample gets a *weight*. For instance, we might count the first value in the data set four times compared with the other only counting once.

As an example, in baseball, the batting average is the number of hits over the number of at-bats. In this calculation, a hit is equal to any other hit, a single is the same as a home run.

On the other hand, the slugging percentage is the total bases divided by the number of at-bats. In this calculation, a home run counts four times as much as a single.

Sample Median

Sample Median

The **sample median** is the “middle number” of a set of data, ranked from smallest to largest.

The key to the median is that *about* half of the data values are above it, and *about* half are below it.

Sample Median

While there is a “formula” to calculate the median, we will use the following algorithm:

1. Rank the values from lowest to highest.
2. If the sample size is odd, the median is the middle ranked value.
3. If the sample size is even, the median is the mean of the two middle ranked values.

Sample Median

Some notes on the sample median:

- Every data *rank* (not value) is included in the calculation.
- Every data *rank* (not value) counts the same in the calculation.
- The order *does* matter in the calculation (eventually).

What do these tell us about how the median compares to the mean?

Note: In *contrast* to the mean, **the median is robust to outliers.**

Sample Median Example

Let's calculate the median salary of our Innkeeper's Coffee customers. Here are the salaries (in thousands of dollars):

32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1

By hand:

First, we put the sample in order:

29.1, 29.6, 32.3, 56.1, 57.0, 64.3, 78.9

Since $n = 7$ is odd, the value at position $n+1/2 = 4$ the median is:

$$\tilde{x} = 56.1$$

In R:

```
salaries <- c(32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1)
median(salaries)
```

Sample Median Example

Now let us include Elon in our sample.

32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1, 20690240.0

By hand:

First, we put the sample in order:

29.1, 29.6, 32.3, 56.1, 57.0, 64.3, 78.9, 20690240.0

Now, because $n = 8$ is even, the sample median is the mean of the middle two digits:

$$\tilde{x} = \frac{56.1 + 57.0}{2} = 56.55$$

In R:

```
salaries2 <- c(32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1, 20690240.0)
median(salaries2)
```

Sample Median Example

Thus, with the inclusion of that single outlier, the median only increased by 0.45.

Because of this, we can see that the median is robust to outliers.

Measures of Position

If we care about describing the center of a variable, we would use a “measure of center”. However, if we care about some other position in the variable, we would use a **measure of position**. Note that the measures of center are a *specific type* of measure of position. Also note that most of these are related to the median instead of the mean.

- Percentiles
- Quartiles
- Quintiles
- Deciles
- z -score (z -transformation)

Quantiles (a.k.a. Percentiles)

The k^{th} **percentile** is the value that separates the lower $k\%$ of the data from the upper $(100 - k)\%$ of the data.

Quantiles are useful when we want to discuss things that are not the middle. For instance:

- In inequality research, we tend to focus on the 10^{th} percentile to understand poverty.
- In basic statistics, we tend to focus on the 2.5^{th} and 97.5^{th} percentiles to better understand uncertainty in our estimates.
- Education research tends to focus on the 5^{th} , 10^{th} , or 90^{th} percentile.

It all depends on your field of study. Ask around in your home department to see which percentile(s) you should focus on.

Quartiles

The median separates the data into two equal parts. There may be times when we wish to divide the ordered data into *four* parts...into “quarters”. The values that divide the *ordered* data into four parts are called the **quartiles**.

- The **first quartile**, Q_1 :

A value that separates the bottom 25% of the data from the top 75%.

- The **second quartile**, Q_2 :

A value that separates the bottom 50% of the data from the top 50%.

- The **third quartile**, Q_3 : A value that separates the bottom 75% of the data from the top 25%.

Note that the **second quartile** is just the median; the value that separates the bottom 50% from the rest.

Other -iles

All of the -iles can be calculated in terms of the percentiles.

- The median is the 50th percentile.
- The values of Q_1 and Q_3 are the 25th and 75th percentiles.
- The quintiles are the 0th, 20th, 40th, 60th, 80th, and 100th percentiles.
- The deciles are the 0th, 10th, 20th, ..., and 100th percentiles

The R function to calculate the percentiles is **quantile** function. The following code calculates the 5th and 95th percentiles of the **salaries** data loaded earlier:

```
quantile(salaries, c(0.05,0.95))
```

Note that the first slot is the variable name and the second is the percentile(s) that needs to be calculated.

Examples

For the original salary data, calculate the following items:

- mean
- first quartile, Q_1
- the 90th percentile
- all quintiles
- the difference between Q_1 and Q_3

Code:

```
salaries <- c(32.3, 56.1, 29.6, 78.9, 64.3, 57.0, 29.1)

mean(salaries)
quantile(salaries, 0.25)
quantile(salaries, 0.90)
quantile(salaries, c(0.0, 0.2, 0.4, 0.6, 0.8, 1.0))
quantile(salaries, 0.75) - quantile(salaries, 0.25)
```

Measures of Spread

Measure of Spread

A **measure of spread** of data describes how similar or varied the set of observed values are for a particular data set. It is used to describe the variability in a sample or population.

Examples:

- standard deviation
- variance
- coefficient of variation
- range
- interquartile range (IQR)

Variance, s^2

Variance

The **variance**, s^2 , is an average (positive) distance to the center of the data set.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Variance, s^2

Some notes on the variance:

- Since the calculation requires using the mean, the variance should only be used if the mean is meaningful.
- Mathematicians like this because variances add.
- The units of the variance is *units*².
- This is the *sample variance*. The *population variance* is denoted as σ^2 .

The R function to calculate the variance is `var()`.

Standard Deviation, s

Standard Deviation

The **standard deviation**, s , is an average (positive) distance to the center of the data set. Specifically, it's the square root of the variance.

$$s = \sqrt{s^2} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Standard Deviation, s

Some notes on the standard deviation:

- Since the calculation requires using the mean, the standard deviation should also only be used if the mean is meaningful.
- The standard deviation is usually preferred when conveying information, because it has the same units as the data.
- This is the *sample standard deviation*. The *population standard deviation* is denoted as σ .

The R function to calculate the standard deviation is `sd()`.

Mean Absolute Deviation, MAD

Variance

The **mean absolute deviation**, MAD , is another measure of the average (positive) distance to the center of the data set.

$$MAD = \frac{\sum_i |x_i - \bar{x}|}{n}$$

The Basic Categorical Graphics

Some basic **univariate** categorical graphics consist of:

- Pie charts
- Bar charts

Some basic **bivariate** categorical graphics consist of:

- Side-by-side bar charts
- Stacked bar charts
- Spine charts
- Mosaic plots

Pie Chart

You're all familiar with pie charts. It is a circle split into segments of the size of the proportion of the sample in each category.

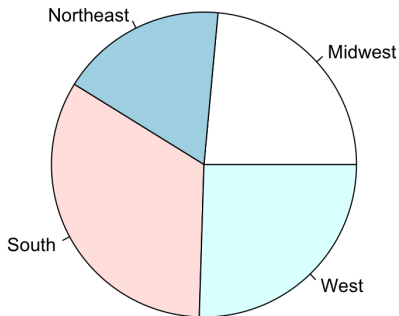
Be careful with pie charts. They can be hard to read and easy to misuse.

The function in R to make a pie chart is `pie(table(var))`.

The `table` function returns the count of each element in a vector.

Pie Chart

```
pie(table(census4))
```



Bar Chart

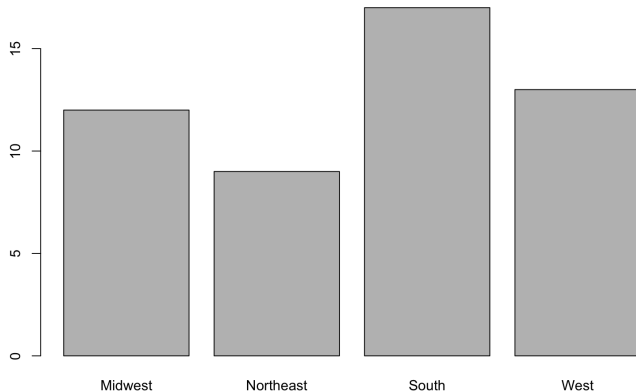
You know bar charts as well. It is a graph with bars whose height represents the frequency of each category.

Often bar charts The function in R to make a bar chart is

`barplot(table(var)).`

Bar Chart

```
barplot(table(census4))
```



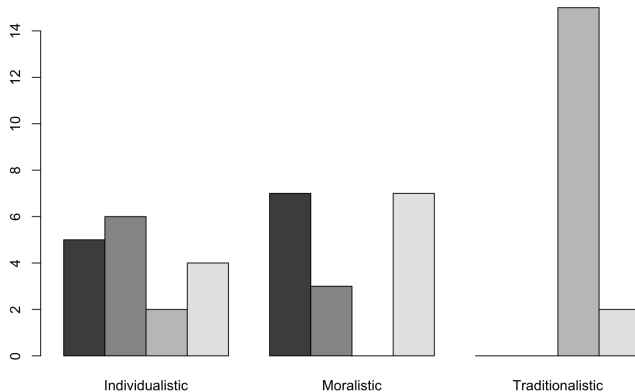
Side-by-Side Bar Chart

A side-by-side bar chart essentially makes a bar chart of one variable for each category of the other variable.

The function in R to make a side-by-side bar chart is `barplot(table(var1, var2), beside=TRUE)`.

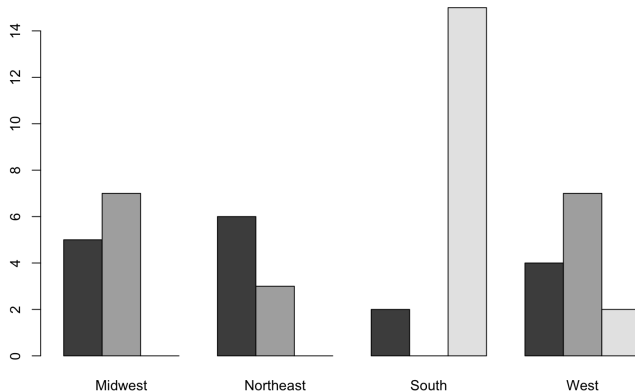
Side-by-Side Bar Chart

```
barplot(table(census4, domPolCulture), beside=TRUE)
```



Side-by-Side Bar Chart

```
barplot(table(domPolCulture, census4), beside=TRUE)
```



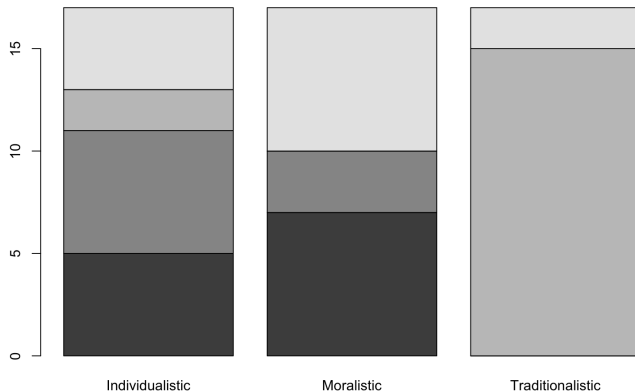
Stacked Bar Chart

A side-by-side bar chart makes a bar chart of one variable and colors in the segments of those bars according to the distribution of the other variable.

The function in R to make a side-by-side bar chart is `barplot(table(var1, var2))`.

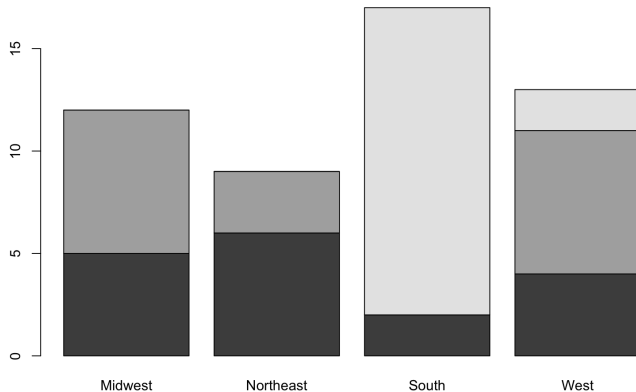
Stacked Bar Chart

```
barplot(table(census4, domPolCulture))
```



Stacked Bar Chart

```
barplot(table(domPolCulture, census4))
```



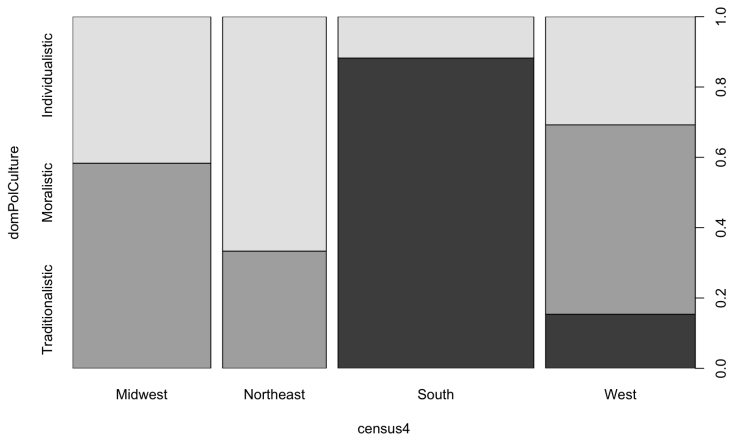
Spine Chart

A spine chart is a stacked bar chart where instead of the height of the bar reflecting the frequency, but the width.

The function in R to make a side-by-side bar chart is `spineplot(table(var1, var2))`.

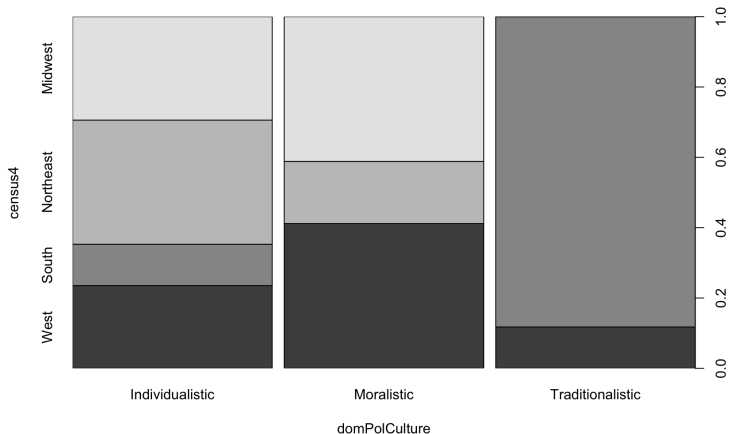
Spine Chart

```
spineplot(table(census4, domPolCulture))
```



Spine Chart

```
spineplot(table(domPolCulture, census4))
```



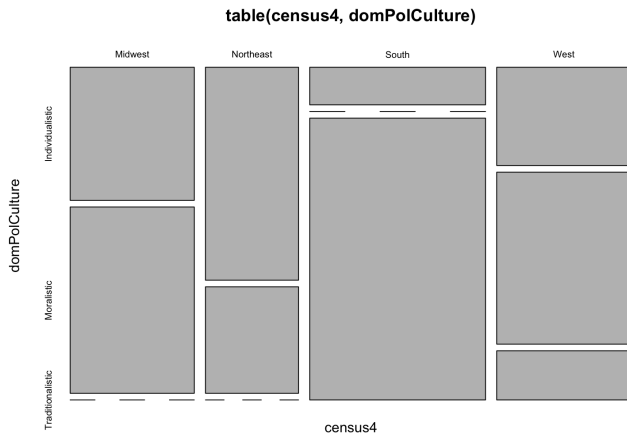
Mosaic Chart

A mosaic chart does the same as the spine chart, but has separate boxes for the categories in the bars.

The function in R to make a side-by-side bar chart is `mosaicplot(table(var1, var2))`.

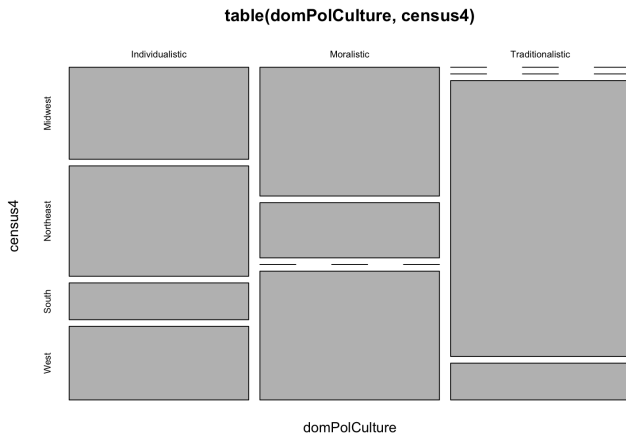
Mosaic Chart

```
mosaicplot(table(census4, domPolCulture))
```



Mosaic Chart

```
mosaicplot(table(domPolCulture, census4))
```



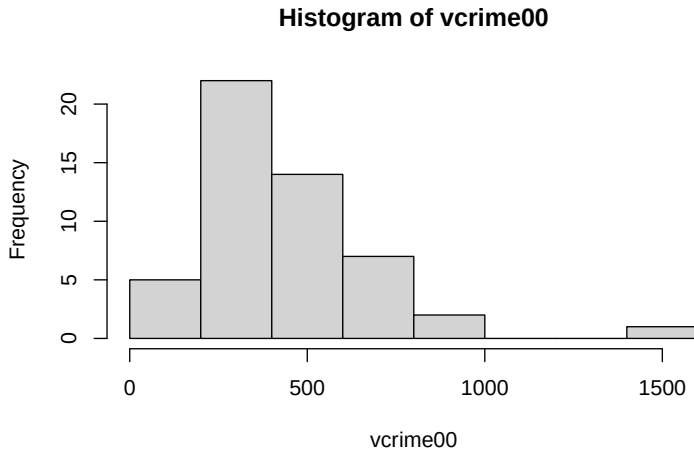
Univariate Graphics for Numeric Graphics

The **univariate** graphics we will discuss are:

- histogram
- density plot
- overlay plot
- box-and-whiskers plot

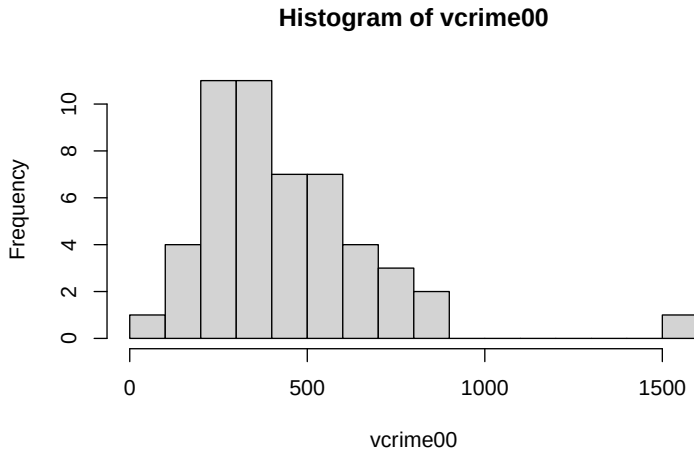
Histogram

```
hist(vcrime00)
```



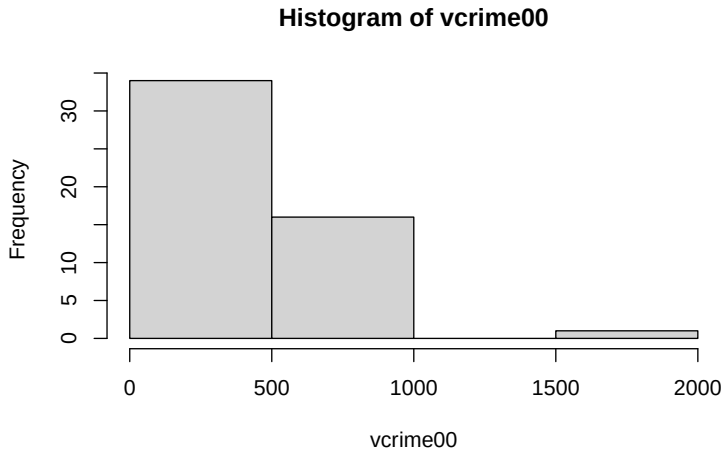
Histogram

```
hist(vcrime00, breaks=11)
```



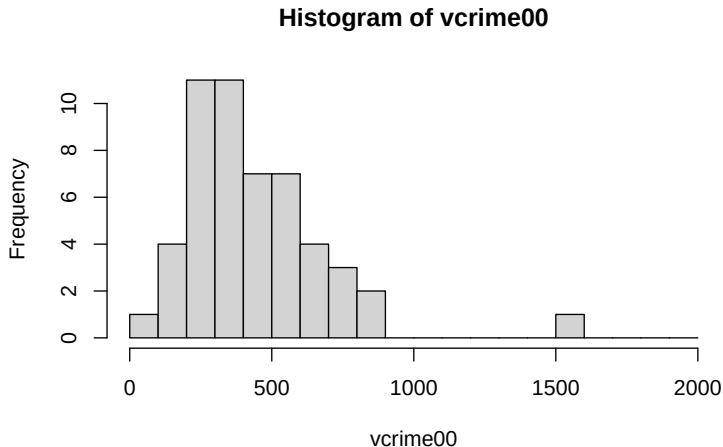
Histogram

```
hist(vcrime00, breaks=5)
```



Histogram

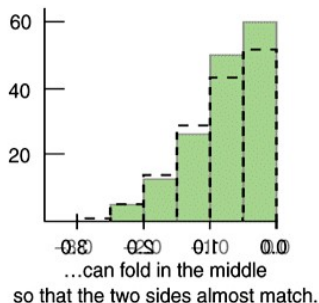
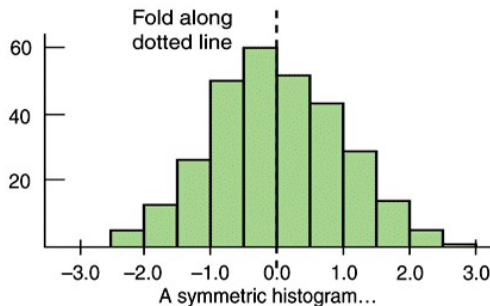
```
hist(vcrime00, breaks=seq(0,2000,100))
```



Shape of a Histogram

A **symmetric** distribution is one which looks the same on both sides if split in half.

This is approximate; no real life distributions will ever be perfectly symmetric.



Shape of a Histogram

If a distribution is not symmetric, we call that distribution **skewed**.

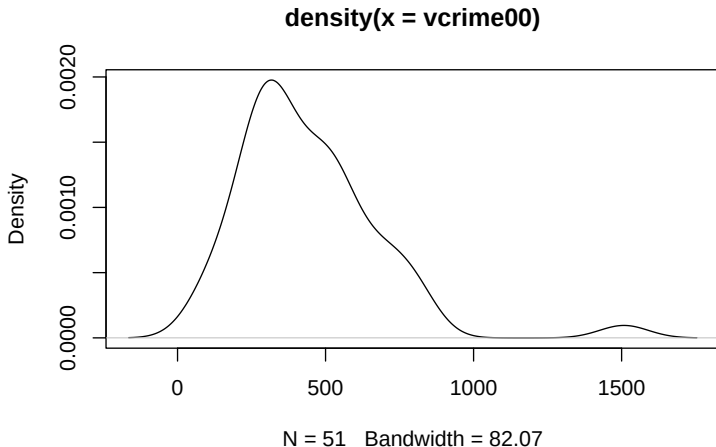
Skewed distributions have long “tails” extending out in one direction.

If the tail is extending out to the left, we say the distribution is **left (or negatively) skewed**.

If the tail is extending out to the right, we say the distribution is **right (or positively) skewed**.

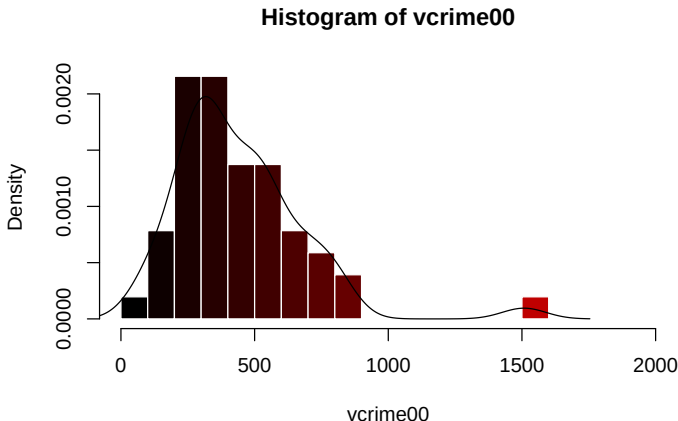
Density Plot

```
plot(density(vcrime00))
```



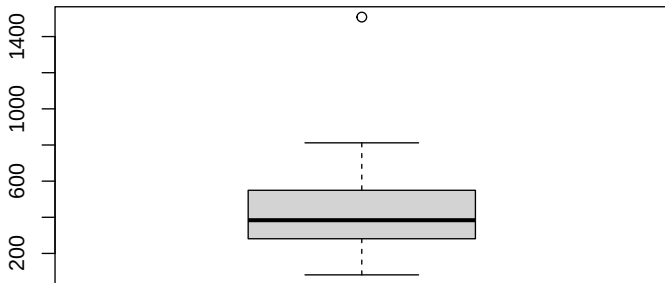
Density Plot

```
hist(vcrime00, freq=FALSE, breaks=seq(0,2000,100),  
col=rgb(0:20/20,0,0), border="white")  
lines(density(vcrime00))
```



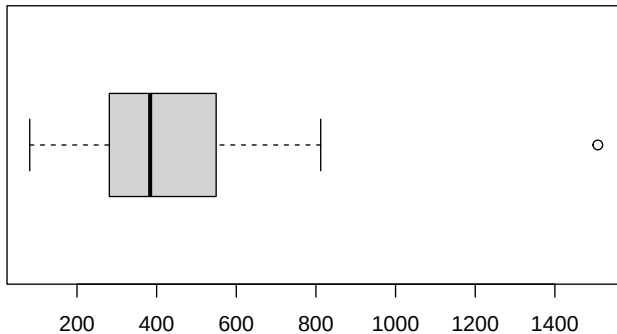
Density Plot

```
boxplot(vcrime00)
```



Density Plot

```
boxplot(vcrime00, horizontal=TRUE)
```



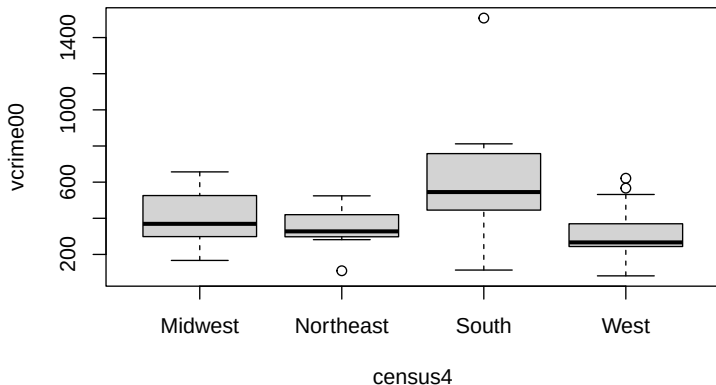
Bivariate Graphics for Numeric Graphics

The **bivariate** graphics we will discuss are:

- side-by-side boxplot
- scatter plot
- line plot

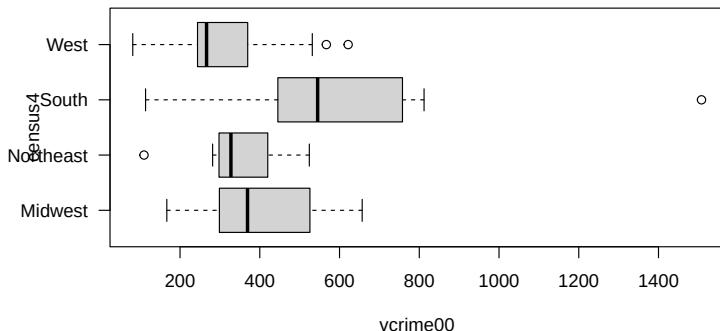
Side-by-Side Boxplots

```
boxplot(vcrime00 ~ census4)
```



Side-by-Side Boxplots

```
boxplot(vcrime00 ~ census4, horizontal=TRUE, las=1)
```



Scatter Plot

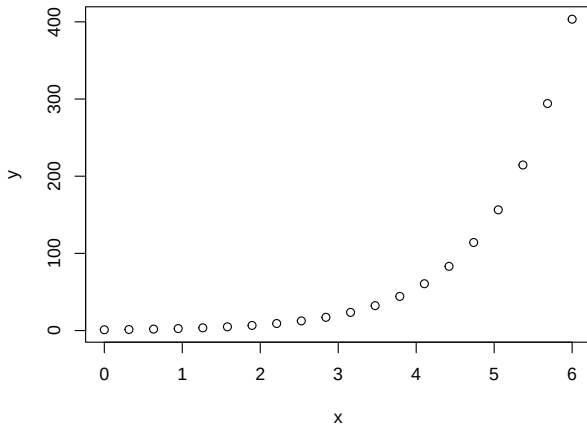
We'll start by making a couple of vectors to plot:

```
# generate 20 evenly spaced numbers between 0 and 6
x <- seq(0, 6, length.out=20)

# create a new vector of  $e^x$  for each x
y <- exp(x)
```

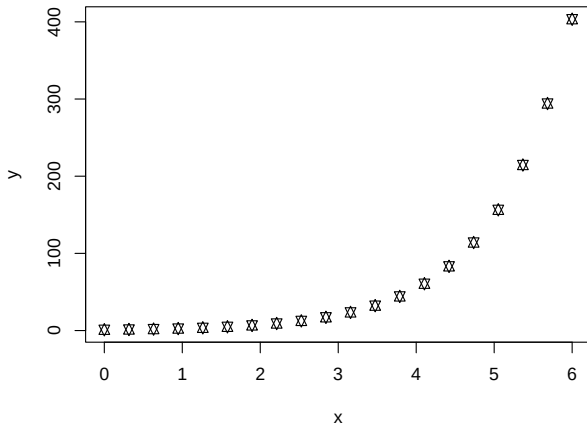
Plots in R

`plot(x,y)`



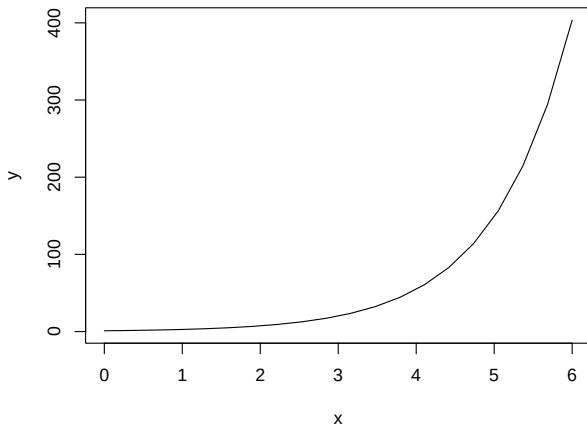
Plots in R

```
plot(x,y, pch=11)
```



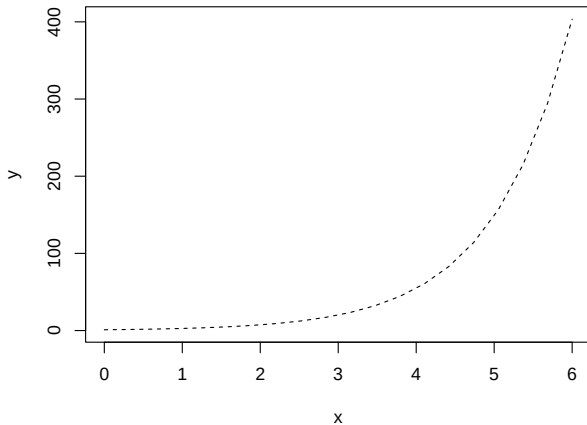
Plots in R

```
plot(x,y, type="l")
```



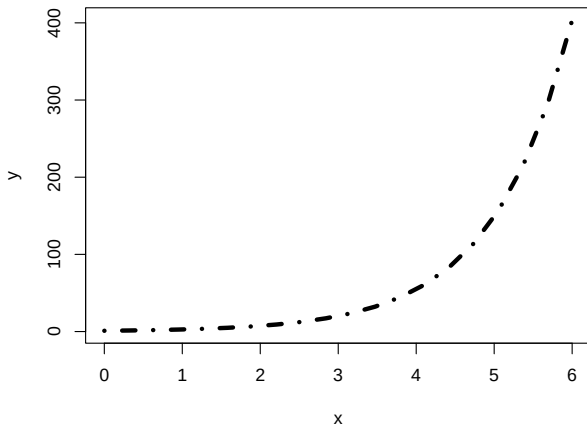
Plots in R

```
plot(x,y, type="l", lty="dashed")
```



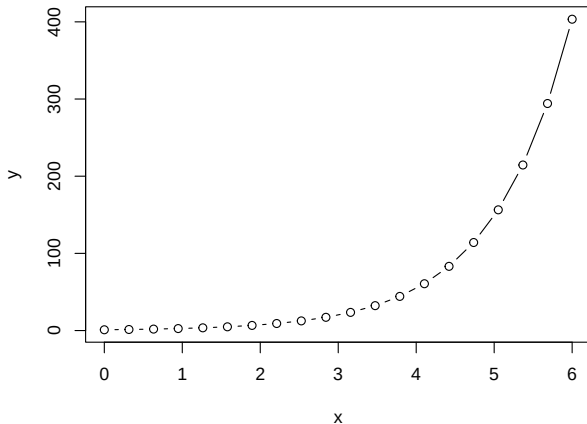
Plots in R

```
plot(x,y, type="l", lty="dotdash", lwd=4)
```



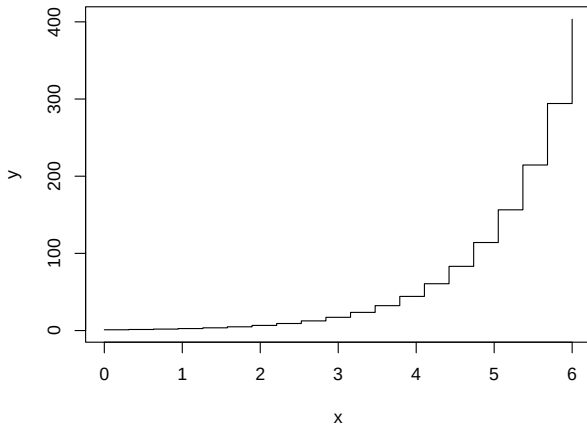
Plots in R

```
plot(x,y, type="b")
```



Plots in R

```
plot(x,y, type="s")
```



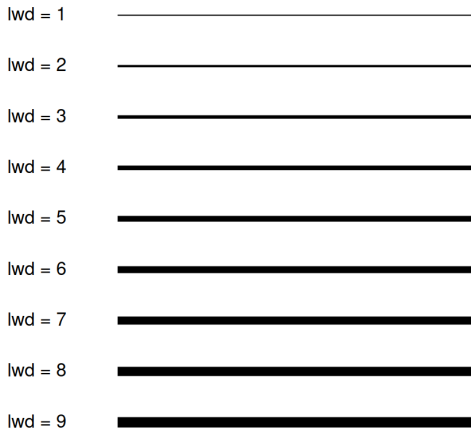
R Point Types (pch =)

0 	1 	2 	3 	4 	
5 	6 	7 	8 	9 	
10 	11 	12 	13 	14 	
15 	16 	17 	18 	19 	
20 	21 	22 	23 	24 	25 

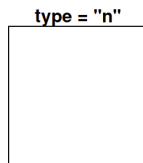
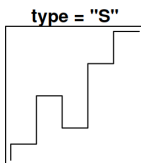
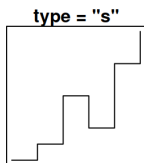
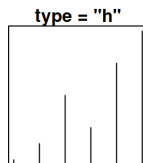
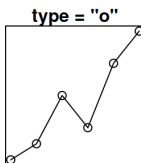
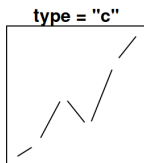
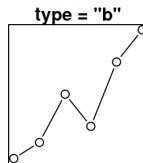
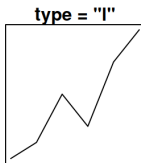
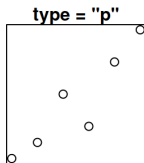
R Line Types (lty =)

0 "blank"		"aa"	
1 "solid"		"1342"	
2 "dashed"		"44"	
3 "dotted"		"13"	
4 "dotdash"		"1343"	
5 "longdash"		"73"	
6 "twodash"		"2262"	

R Line Widths (`lwd =`)

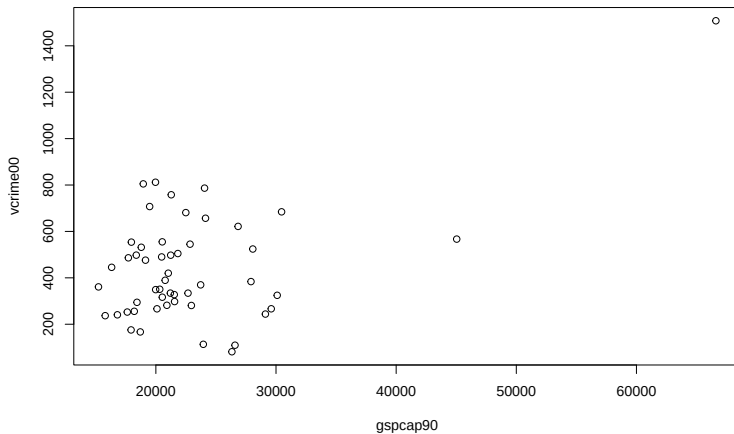


R Plot Types (`type =`)



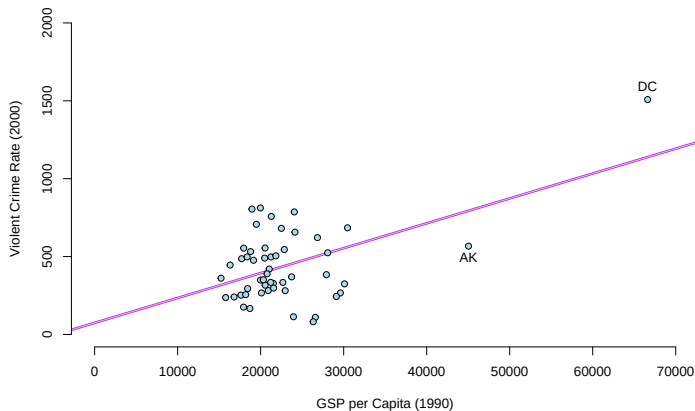
Scatter Plot

```
plot(vcrime00 ~ census4)
```



How Do We Make This Plot?

Suppose we want to make a plot like this. How can we make something this fancy?



The Parameters

The following code:

- sets the margins to be 4, 4, 1, and 1 (lines at the bottom, left, top, and right);
- specifies the font family is serif, with labels bolded;
- specifies the labels are 20% larger and values are 20% smaller than default;
- requires the axes to meet at 0 (no axis padding); and
- forces all labels to be horizontal (for the sake of your readers).

```
par(mar=c(4,4,1,1))  
par(family="serif", font.lab=2)  
par(cex.lab=1.2, cex.axis=0.8)  
par(xaxs="i", yaxs="i")  
par(las=1)
```

The Start

The following code:

- starts the graphic; and
- sets the viewing window to $(0, 70000) \times (0, 2000)$.

```
plot.new()  
plot.window(xlim=c(0,70000), ylim=c(0,2000))
```

Note that this will start a generic window of that size/dimension.

Again, some graphing functions have their own “starting” functions. So, in those cases, the above two lines are not needed.

The Annotation

The following code:

- adds the line of best fit (looking all sexy);
- adds the points (light-blue-filled circles);
- adds the values and labels; and
- identifies the two outliers.

```
abline(lm(vcrime00~gspcap90), lwd=3, col="purple")
```

```
abline(lm(vcrime00~gspcap90), lwd=1, col="pink")
```

```
points(gspcap90 ,vccrime00 , pch=21, bg="lightblue")
```

```
axis (1); axis (2)
```

```
title(xlab="GSP per Capita (1990)")
```

```
title(ylab="Violent Crime Rate (2000)")
```

```
text(gspcap90[24],vccrime00[24], label=scode[24], pos=3)
```

```
text(gspcap90[50],vccrime00[50], label=scode[50], pos=1)
```

Our Plot

All this comes together to produce:

